From Cycle to Window

Paper 4. What survives when an exact periodic law is evaluated on a short interval

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Abstract

Paper 3 gives exact laws over a full cycle of length $\prod q$. Applications need a window of length about z^2, which is exponentially shorter. This paper asks what survives.

Proved. An old line $q \geq 7$ acts on a new one as a periodic ruler, and in a first-cycle sector it strikes either once or not at all (Theorems 1 and 2). The deficit telescopes to one sum (Theorem 3):

$$T(U) - M(U) = \sum_{r \leq U} B_r(U).$$

On the full cycle the survivor masks form a martingale with energy $\sum \mathbf{E}(\Delta_r^2) = 1/P_z - 1$ (Theorem 4). Over a full cycle of sectors the boundary errors of any set of lines sum to $2^{|S|} - \prod_{r \in S}(2 + (2/r))$ (identity B), and the mean deviation of a line carries the Legendre symbol $(2/r)$ (identity A); both are proved in §4.

Measured. On every-integer sectors at moving depth, the ratio of twins to the sieve product is

$$T(U)/M(U) = 0.793733 \quad \text{at } U = 10^5,$$

against the limit $e^{2\gamma}/4 = 0.7930547$ that the Hardy–Littlewood conjecture predicts. Split by line depth, the deficit follows a Buchstab model to within 0.0007.

Corrected. The L^2 bound (5.1) set as the target in earlier versions would force $T/M \to 1$, so it cannot hold together with the conjecture. It is replaced by a sufficient condition: $T \geq M(1 - \sqrt{Q})$, with $Q(U) = k(U)E(U)/M(U)^2$. The measured Q is below 0.1 but rising; no bound on it is proved.

Transfer (§6). Soft depth weights transfer to the window at ratio near 1; the sharp depth-zero indicator, which is the twin condition, transfers at about 0.80 (Proposition 1). Patterns of up to five cells transfer at 1.0000 (Proposition 2), which puts the loss in the one-point density.

Keywords: Buchstab function, sieve of dimension two, martingale decomposition, large sieve, weighted sieves.

MSC 2020: 11N35, 11N05, 11K38.

How to read the claims in this paper. Statements set as Theorems, Propositions and Corollaries are proved, and the proofs are given. Everything else falls into two kinds, and we try to keep them apart. A *measurement* is a computation over a stated finite range; it is labelled with that range, and it supports a claim about that range only. A *reading* is our own judgement about what a measurement or a proof appears to mean, and we mark it as ours rather than stating it as established.

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1. Setting, and what is imported

Coordinates from Paper 2: $L_p(k) = p(p + 2k)$; L_3 as the grid; cells $C_b = (6b - 1, 6b + 1)$; sectors bounded by consecutive odd squares. Window combinatorics from Paper 1: the increments W_j of $\lfloor 2j^2/p \rfloor$, with $H_j = 2 + W_j$. Exact cycle laws from Paper 3: the four-state law and its refinement by inheritance depth, with generating function $\prod_q((q-2) + xu + xv)$.

Two imported facts do real work below and are stated once:

- **[P1, Thm 1].** Within the first cycle, a sector of p's own row — between $(p + 2j)^2$ and $(p + 2j + 2)^2$ with $j \le p - 1$ — contains between two and six strikes of L_p, for every odd p: the uniform bound on the increments of $\lfloor 2j^2/p \rfloor$. This is what makes §2.3 a *binary* statement for old rulers $q \ge 7$; L_5 is handled separately as part of the fixed core.
- **[P3, Thm 2].** On the full cycle, each line contributes nothing with probability $(q-2)/q$, a lower-rail strike with probability $1/q$ and an upper-rail strike with probability $1/q$, independently across lines. This is the exact reference distribution against which every window measurement below is compared.

The comparison object throughout is a ratio: a quantity measured on the window, divided by the same quantity computed from [P3, Thm 2] on the cycle. A ratio of 1 means the law transfers.

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2. The strip, and old lines as rulers

2.1 The strip

A line is nothing but the strip $k = 0, 1, 2, \ldots$, and the squares do not alter it — they cut it into blocks. For L_7:

sector	k	H
$(49, 81]$	1, 2	2
$(81, 121]$	3, 4, 5	3
$(121, 169]$	6, 7, 8	3
$(169, 225]$	9–12	4
$(225, 289]$	13–17	5
$(289, 361]$	18–22	5
$(361, 441]$	23–28	6

The blocks are contiguous (verified over 20 sectors): the squares never reset a line's phase, so a bias in one sector is continued, not restarted, by the next.

2.2 Theorem 1 (old lines as rulers)

> **Theorem 1.** For a new line p and an old line $q < p$, with $x = L_p(k)$, $x + 6j \equiv 0 \pmod q \iff k \equiv k_0 - jd \pmod q$, $\quad d = 3p^{-1} \bmod q$, where $k_0 = -p^2(2p)^{-1} \bmod q$.

Proof. $x + 6j = p^2 + 2pk + 6j \equiv 0$ gives $k \equiv (-p^2 - 6j)(2p)^{-1} = k_0 - 3jp^{-1}$. ■

Verification. Zero failures among 35,868 instances.

The coefficient is 3, not 6, because one step in k moves the integer by $2p$; matching a displacement of 6 modulo q requires $2pd \equiv 6 \pmod{q}$.

Equivalently, in clock form. The first inherited strike index is

$$\phi_q(p) \equiv \frac{q-p}{2} \pmod{q}, \qquad (2.2)$$

zero failures among 4,983 instances.

2.3 Theorem 2 (binary decision) and phase collisions

Let $\kappa_j = 2j + \lfloor 2j^2/p \rfloor$ be the last k-index at the left square boundary, as in [P2, §0.1]. For an old ruler q, define the distance from the start of the block to its next mark by

$$\delta_q(j) = 1 + ((\phi_q(p) - \kappa_j - 1) \bmod q) \in \{1, \ldots, q\}.$$

Thus the block contains a mark of q exactly when this next-mark distance does not exceed the block length.

> **Theorem 2.** For every old line $q \geq 7$ and every sector j **in the first cycle** $0 \leq j \leq p-1$, the number of strikes of q inside the block is $C_q(j) \in \{0, 1\}$; and $C_q(j) = 1$ exactly when $\delta_q(j) \leq H_j$, in which case the strike falls in slot $s = \delta_q(j)$.

Proof. By [P2, §0.1] and [P1, Thm 1], $H_j = 2 + W_j \leq 6 < 7 \leq q$ throughout the first cycle, so a block is shorter than the ruler's period and cannot contain two of its marks. ■

Two hypotheses, both necessary.

- *The first cycle.* Beyond it $H_j = H_j^{(0)} + 4r$, so a block can be longer than the ruler's period and the conclusion fails. Counterexamples at $p = 11$, $q = 7$: the sectors $j = 18, 21, 23, 25, 27, 28, 29, 30, 31, 32$ (with $H_j = 9$ to 14) each contain **two** marks of 7 — ten such sectors in the first three cycles. Every use of Theorem 2 below is inside the first cycle.
- *The ruler $q \geq 7$.* The ruler $q = 5$ can meet a block of length 6 twice. In the moving-depth calculations L_5 is included explicitly in the fixed core; all subsequent uses of the binary ruler statement refer to $q \geq 7$.

For those rulers the local question changes from counting to presence/absence, and the state of a block is a **word** W_j of length $H_j \leq 6$.

Composites are phase collisions. If $\delta_{q_1}(j) = \delta_{q_2}(j) = s$, both lines mark slot s, i.e. the integer there carries both factors. Verified at $p = 11$, sector 2: the cofactors $21, 23, 25$ give $O_{\{3,7\}}, N, O_{\{5\}}$ — and $21 = 3 \times 7$ appears precisely because the two rulers' phases coincided.

3. Moving depth

3.1 Setup

Let the depth track the window: sieve each sector (u^2, v^2) by all lines $\leq u$, and set

Here u runs over the integers $\equiv 5 \pmod 6$ and $v = u + 2$, which is then also coprime to 6; a **survivor** is a cell with both members surviving, $C(u)$ is the number of cells of the sector, and $P(u) = \prod_{5 \leq q \leq u}(1 - 2/q)$ as in Theorem 3 below. Note what this makes T: since every odd composite

below v^2 has a prime factor at most u, sieving to u leaves exactly the twin pairs, so T counts twins rather than estimating them. **Section 5 indexes its sectors differently** — see there.

$$T(U) = \sum_{u \le U} (\text{survivors}), \qquad M(U) = \sum_{u \le U} C(u)P(u).$$

3.2 The measured ratio

range of n	2–50	51–150	151–300	301–600	601–1000	1001–1600
T/M	0.86709	0.82736	0.79313	0.84748	0.81585	0.80150

Cumulatively over the whole range, $T/M = 0.81152$.

$$\boxed{T/M \approx 0.80, \text{ across three orders of magnitude.}}$$

(Regenerated by `code/verify_l2_stress.py` from the definitions of §3.1. The per-bin values fluctuate more than a cumulative reading would; the stable quantity is the cumulative ratio.)

The discrepancy is systematic, not numerical noise. At moving depth the sieve variable is $s = \log(u^2)/\log u = 2$ exactly, the critical scale at which a naive product is expected to need a Buchstab-type correction [2]. For comparison, in one dimension the familiar factor at $s = 2$ is $e^\gamma/2 = 0.8905362$; in the present two-rail experiment the corresponding measured ratio is about 0.80.

What the ratio should tend to. Take the every-integer sectors $(n^2, (n+1)^2)$ of §5, and let C_2 be the twin-prime constant. Mertens' product gives $P(n) \sim 12C_2e^{-2\gamma}/\log^2 n$, so $M(U) \sim 2C_2e^{-2\gamma}U^2/\log^2 U$. The Hardy–Littlewood conjecture [3] gives $T(U) \sim (C_2/2)U^2/\log^2 U$. If that conjecture holds, then

$$\frac{T(U)}{M(U)} \to \frac{e^{2\gamma}}{4} = 0.7930547.$$

This limit is conditional. Measured on those sectors, the ratio is 0.810480 at $U = 300$, 0.795771 at $U = 10^4$ and 0.793733 at $U = 10^5$. So a ratio near 0.80 is what the conjecture predicts; on this reading it is the expected main-term correction, not a loss of twins.

Where the loss sits, by depth. Theorem 3 below splits $T - M$ into one term for each pair (sector n, line p). Put each term in a bin by the depth $a = \log p/\log n$ of the line in its sector, and add the bins up to a_0. For one sector this partial sum is exactly $P(n)(N_z(n)/P(z) - C(n))$ with $z = n^{a_0}$, where $N_z(n)$ counts the cells that survive the lines up to z. Divided by M, the measured curve is compared with

$$\left(e^\gamma \omega(2/a_0)\right)^2 - 1,$$

where ω is Buchstab's function. This is the curve the two ends of a cell would give if each behaved like an independent integer with no prime factor below n^{a_0}. It is a heuristic model, and at $a_0 = 1$ it is the conditional limit above.

a_0	0.5	0.6	0.7	0.8	0.9	1.0
measured, $U = 10^5$	−0.00007	−0.00131	+0.01830	+0.00204	−0.07410	−0.20627
model	0.00000	−0.00167	+0.01862	+0.00259	−0.07395	−0.20695

Over all twenty bins, the largest gap between the two rows is 0.0007 at $U = 10^5$ and 0.0013 at $U = 3 \times 10^4$. The lines with $a \le 0.5$ add about nothing. Between $a = 0.6$ and $a = 0.7$ the sum rises to about +1.8 per cent, and by $a = 0.8$ it has fallen back to about +0.2 per cent. So most of the net loss comes from lines deeper than $a = 0.8$. *(Regenerated by* `code/verify_depth_profile.py`*; the model is checked against the data, not proved.)*

A struck endpoint near the top has a simple shape. Suppose $x = pt$ is struck by p in the sector n and its cell survived every smaller line. Then t has no prime factor below p. If $p^3 > (n+1)^2$, then $t < p^2$, so t is prime. The weaker condition $a > 2/3$ is not enough in a finite sector: at $n = 36$, $p = 11$ the cell $(1331, 1333)$ has $a = 0.669$ and $t = 121$.

3.3 Why accumulation matters more than a single sector

One does not need $T(u) > 0$ for every sector, only $T(U) \to \infty$. On the indexing of this section — $u \equiv 5 \pmod 6$ with the window $(u^2, (u+2)^2)$, so that only one sector in six is swept — one has $\sum_{u \le U} C(u) \asymp U^2/18$, against $U^2/6$ for the every-integer indexing of §5; the two sums differ by that factor of three and only the constant is affected. With $P(U) \asymp 1/\log^2 U$,

$$M(U) \asymp \frac{U^2}{\log^2 U}, \qquad (3.1)$$

in either indexing, so **any error bound of size $O(U \log^A U)$ suffices** — a requirement weaker by a full factor of U than square-root cancellation. This is a sufficient condition only. Under the Hardy–Littlewood conjecture no such bound holds: $T - M$ is then close to $-M/5$, of the same order as M (§3.2, §5.1). What can be hoped for is a bound on the error *relative to* M with a constant below 1, as in §5.1.

3.4 Theorem 3 (telescoping to a single sum)

The three symbols the theorem uses, defined here. Inside the sector at u, write $N_{r^-}(u)$ for the number of cells still open after every line below r has acted, and $D_r(u)$ for the number of those that the line r then closes — a strike of r on a survivor of the smaller lines, not a raw strike. The **local deviation** is the difference between that and the cyclic prediction $2/r$ of [P3, Thm 2]:

$$\varepsilon_r(u) = D_r(u) - \frac{2N_{r^-}(u)}{r}. \qquad (3.2)$$

Two facts about it are used below and were established in the coordinates of [P2, §4.5]: inside a sector the line r deletes at $s \equiv -u^2/6$ and $s \equiv (2-u^2)/6 \pmod r$, so the two positions are 3^{-1} mod r apart (zero failures among 85 tested); and **every ruler's phase is therefore a function of u^2 alone**, so under $u \mapsto u + 6$ it moves quadratically, $u^2 \mapsto u^2 + 12u + 36$. That quadratic motion is what makes cancellation possible at all. The superscript form $D_r^{(0)}(n)$ used in §4.2 is the same count taken with no earlier lines acting, i.e. the raw strike count of r in the sector n.

Theorem 3. With $P(r) = \prod_{5\le q\le r}(1-2/q)$, $T(u) = P(u)\left[C(u) - \sum_{r\le u}\dfrac{\varepsilon_r(u)}{P(r)}\right]$, $T(U) - M(U) = \sum_{r\le U} B_r(U)$, where $B_r(U) = -P(r)^{-1}\sum_{u\ge r} P(u)\varepsilon_r(u)$.

Proof. From $N_r = N_{r^-}(1-2/r) - \varepsilon_r$, divide by $P(r)$ and telescope. ■

Verification. At $U = 1019$: $T = 2920.0$, $M = 3535.3$, $T - M = -615.3 = \sum B_r$. At $U = 2039$: -2143.2 on both sides.

$2^{\pi(U)}$ intersections $\longrightarrow$ one sum over primes.

This is the first point in the framework at which stacking the lines does not blow up.

3.5 The state space is the cycle

Define the clock vector here by $\Phi_z(p) = (\phi_3(p), \phi_5(p), \ldots, \phi_z(p))$, using (2.2) componentwise. It takes values in a set of size $\prod_{3\le q\le z} q$, while a trajectory of length N visits at most N of them:

z	lines	states $\prod q$
13	5	15,015
31	10	1.0×10^{11}
59	16	9.6×10^{20}
101	25	1.2×10^{38}

The survivor pattern of the lines up to z has period exactly $\prod q$ and, by [P3, Thm 1], exactly $\prod(q-2)$ surviving cells in it. **Any "compressed law for the next survivor" — a rule carrying a state materially smaller than the cycle — would therefore be a compression of the cycle itself.** We record this as an observation rather than a theorem, but it is the reason any "successor tower" for the sieved strip is a re-description and not a reduction.

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4. Decomposing the deviations

4.1 Identity A (the mean, and its Legendre symbol)

Split the deviation (3.2) as $\varepsilon_r = \mu_r + \xi_r$ into mean and fluctuation, the mean μ_r being taken over a full period: $\prod_{5\le q\le r} q$ consecutive sectors of the family of §3, all with $u \ge r$.

Identity A. $\mu_r = -\dfrac{\chi_r}{r} Q_{r^-}$, where $\chi_r = \left(\frac{2}{r}\right)$ and $Q_r = \prod_{5\le q\le r}\left(1 - \frac{2+\chi_q}{q}\right)$.

The proof rests on identity B and is given after it, in §4.2.

Origin of the symbol. At a square boundary the line r can strike when $u^2 \equiv 0$ or $u^2 \equiv 2 \pmod r$; the number of admissible phases of u is therefore $1 + (1 + \chi_r) = 2 + \chi_r$, and a CRT count suggests the product structure across a set of lines. The proof in §4.2 makes this count exact.

Check. Over 1,400 consecutive sectors (not a whole number of periods) with standard errors, the z-scores against the prediction are $0.04, 0.01, -0.44, 0.26, 0.49, 0.33, -1.03, 0.48, 0.32, -0.80$ for $r = 5, \dots, 101$ — all within one standard error. The two cleanest cases agree to three digits: $r = 5$ predicts 0.200000 and measures 0.200286; $r = 7$ predicts -0.114286 and measures -0.113959.

4.2 Identity B (square-cycle cancellation)

Identity B. With $\eta_r(n) = D_r^{(0)}(n) - 2C(n)/r$, summed over a full cycle of r sectors, $\sum_{\text{cycle}} \eta_r = -\left(\frac{2}{r}\right)$; and for any set S of lines, over a cycle of length $M_S = \prod_{r\in S} r$, $\sum_{\text{cycle}} \eta_S = 2^{|S|} - \prod_{r\in S}\left(2 + \left(\frac{2}{r}\right)\right)$.

Here $D_S^{(0)}(n)$ counts the cells of the sector with a member divisible by every line of S, and $\eta_S(n) = D_S^{(0)}(n) - 2^{|S|}C(n)/M_S$. When $u \geq \max S$ every such multiple lies above the line's square, so this is the raw strike count.

Proof. Take a sector (u^2, v^2) with u, v coprime to 6. Since $u^2 \equiv v^2 \equiv 1 \pmod 6$, its cells are C_m for $L \leq m \leq R$ with $6L = u^2 + 5$ and $6R = v^2 - 7$, so $C = R - L + 1$. Put $M = M_S$. A cell C_m has a member divisible by every $r \in S$ exactly when $6m \equiv \pm 1 \pmod r$ for each r, that is when $(6m)^2 \equiv 1 \pmod M$; this gives $2^{|S|}$ classes ρ of $6m$. Counting the $m \in [L, R]$ in one class gives $C/M + \lambda(u^2 - 1 - \rho) - \lambda(v^2 - 7 - \rho)$, where $\lambda(y) = ((6^{-1}y) \bmod M)/M$. So

$$\eta_S = \sum_{\rho^2 \equiv 1} [\lambda(u^2 - 1 - \rho) - \lambda(v^2 - 7 - \rho)].$$

Over a cycle of M consecutive sectors, u and v each run through all residues modulo M. Let $K(t)$ be the number of pairs (w, ρ) modulo M with $\rho^2 \equiv 1$ and $w^2 - \rho \equiv t$. Writing $y = 6s$, so that $\lambda(y) = s/M$ for $0 \leq s < M$,

$$\sum_{\text{cycle}} \eta_S = \frac{1}{M}\sum_{s=0}^{M-1} s[K(6s+1) - K(6(s+1)+1)] = \frac{1}{M}\sum_t K(t) - K(1) = 2^{|S|} - K(1),$$

by summation by parts, since $6M + 1 \equiv 1$ and $\sum_t K(t) = 2^{|S|}M$. By the Chinese remainder theorem $K(1)$ is a product over $r \in S$, and modulo r the pairs with $w^2 - \rho \equiv 1$ are $\rho = -1, w = 0$ and $\rho = 1, w^2 \equiv 2$: there are $2 + (2/r)$ of them. With one line this is $2 - 2 - (2/r) = -(2/r)$. ■

The proof uses only that u and v each run through all residues modulo M. So the cycle sum does not depend on the starting sector, and it holds for the family $u = 6n - 1$ of §3, for the family $u = 6n + 1$, $v = 6n + 5$, and for windows between consecutive integers coprime to 6, which is their union.

Proof of identity A. Let T run over the sets of lines below r. By inclusion–exclusion, $D_r = \sum_T (-1)^{|T|} D_{T\cup\{r\}}^{(0)}$ and $N_{r^-} = \sum_T (-1)^{|T|} D_T^{(0)}$, with $D_\varnothing^{(0)} = C$. The main terms cancel and

$$\varepsilon_r = \sum_T (-1)^{|T|}\Big(\eta_{T\cup\{r\}} - \frac{2}{r}\eta_T\Big), \qquad \eta_\varnothing = 0.$$

A full period is a whole number of cycles for every T, so identity B gives the mean of η_T as $(2^{|T|} - \Pi_T)/M_T$, with $\Pi_T = \prod_{q\in T}(2 + \chi_q)$. Each T then contributes $-(\chi_r/r)(-1)^{|T|}\Pi_T/M_T$, and the sum over T is the expansion of $-(\chi_r/r)Q_{r^-}$. ■

Checks. The single-line law holds exactly for $r = 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47$. The general law gives $\{5, 7\} \mapsto 1$, $\{5, 11\} \mapsto 3$, $\{7, 17\} \mapsto -5$, $\{7, 11\} \mapsto 1$, $\{11, 13\} \mapsto 3$, $\{5, 13\} \mapsto 3$, $\{5, 7, 11\} \mapsto 5$. Identity A holds exactly over full periods for $r = 5, 7, 11, 13$: $1/5$, $-4/35$, $16/385$, $32/1001$. All are regenerated by `code/verify_proved_identities.py`.

r boundary errors, each of size $O(1)$, collapse to ± 1. Consequently the whole inclusion–exclusion compresses into two products:

$$E_{\text{cycle}} = M \left[\prod_r \left(1 - \frac{2}{r}\right) - \prod_r \left(1 - \frac{2 + (2/r)}{r}\right) \right].$$

The second product is a factor the purely periodic model does not contain; it is contributed by the square clock.

4.3 Algebraic consequence of identity A

By identity A, $\sum_r -\mu_r/P_r = 1 - Q_z/P_z$ (the sum telescopes, since $Q_r/P_r - Q_{r^-}/P_{r^-} = -(\chi_r/r)Q_{r^-}/P_r$), whence

$$E_{\text{mean}}(U) = \sum_{u \le U} (P_u - Q_u) = O\left(\frac{U}{\log^2 U}\right),$$

smaller than the main term (3.1) by a full factor of U.

Check. On windows between squares of consecutive integers coprime to 6 ($u = 5, 7, 11, 13, \ldots$), at $U = 1000$ the total error is -2057 while the mean part contributes -6.93; at $U = 2000$, -6479 against -11.24 (`code/verify_proved_identities.py`). **The means account for 0.17% of the deviation.**

4.4 Theorem 4 (martingale structure)

Normalise the survivor mask by $Z_z = A_z/P_z$ and set $\Delta_r = Z_r - Z_{r^-}$.

> **Theorem 4.** Over a full CRT cycle, $\mathbf{E}(\Delta_r \mid \mathcal{F}_{r^-}) = 0$, $\mathbf{E}(\Delta_r \Delta_s) = 0$ for $r \neq s$, and $\sum_{r \le z} \mathbf{E}(\Delta_r^2) = \dfrac{1}{P_z} - 1$.

Proof. Put the uniform probability measure on one full CRT cycle, and let $\mathcal{F}_{r^-}$ be the sigma-algebra generated by the residue coordinates of the lines below r. Write $A_r = A_{r^-} B_r$, where B_r is the indicator that the r-coordinate avoids the two forbidden rail positions. Conditional on $\mathcal{F}_{r^-}$, the new CRT coordinate is uniform, so

$$\mathbf{E}(B_r \mid \mathcal{F}_{r^-}) = 1 - \frac{2}{r},$$

and it is independent of the earlier coordinates. Since $P_r = P_{r^-}(1 - 2/r)$,

$$\mathbf{E}(Z_r \mid \mathcal{F}_{r^-}) = \mathbf{E}\left(\frac{A_{r^-} B_r}{P_{r^-}(1 - 2/r)} \middle| \mathcal{F}_{r^-} \right) = Z_{r^-}.$$

Hence $\mathbf{E}(\Delta_r \mid \mathcal{F}_{r^-}) = 0$, so the Δ_r are martingale differences and are pairwise orthogonal. Finally $\mathbf{E}(Z_z) = 1$ and, because $A_z \in \{0, 1\}$,

$$\mathbf{E}(Z_z^2) = \frac{P_z}{P_z^2} = \frac{1}{P_z}.$$

Thus $\mathrm{Var}(Z_z) = 1/P_z - 1$, and orthogonality gives $\sum_{r\le z} \mathbf{E}(\Delta_r^2) = 1/P_z - 1$. ■

Verification. For $\{5, 7, 11, 13\}$ with $W = 5005$: the individual energies are

$$0.6666667, \qquad 0.6666667, \qquad 0.5185185, \qquad 0.5185185,$$

summing to **2.3703703704**, while $1/P - 1 = \mathbf{2.3703703704}$ — agreement to ten digits. All six cross-correlations are $\sim 10^{-17}$, i.e. zero to machine precision.

Since $P_z \asymp 1/\log^2 z$, the total L^2 energy of *all* lines is $O(\log^2 z)$, not exponential. **The primorial is present as a period, but the size of the L^2 error does not carry it.**

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5. The decisive L^2 question

5.1 Statement

The sectors here are not those of §3. In this section u runs over **every** integer with the window $(u^2, (u+1)^2)$, again sieved by all lines $\le u$; $X = X(U)$ is the number of cells swept by all the sectors up to U, and $X_r(U)$ the number swept by the sectors $u \ge r$ alone, in which the line r has been born. The deviation is

$$D_r(U) = (\text{all strikes of } r \text{ over the sectors } u \ge r) - \frac{2X_r(U)}{r}.$$

The normalisation matters, and an earlier version of this section used the wrong one. Subtracting $2X/r$ instead of $2X_r/r$ charges the line r with $2/r$ of the cells of the sectors below its own birth, a deterministic quantity of size $2(X - X_r)/r$; summed and squared it dominates everything else and grows like $\sum_{p\le U} p^2 \asymp U^3/\log U$. That term, and not any property of the sieve, is what the table printed in earlier versions was measuring.

By Cauchy–Schwarz, $\left|\sum_{r\le U} B_r\right| \le \sqrt{\pi(U)}\left(\sum_{r\le U} |B_r|^2\right)^{1/2}$ with B_r as in Theorem 3. Hence if

$$\sum_{r\le U} \left|B_r(U)\right|^2 \ll (X + U^2)\mathrm{polylog}(U), \qquad X \asymp U^2, \qquad (5.1)$$

the total error would be $O(U^{3/2}\mathrm{polylog})$ against a main term $\asymp U^2/\log^2 U$.

Correction: (5.1) asks for too much. Earlier versions read (5.1) as the bound that would give $T(U) \to \infty$. It would, but it would also force $T(U)/M(U) \to 1$. The Hardy–Littlewood conjecture predicts the limit $e^{2\gamma}/4 = 0.7930547$ instead (§3.2). So (5.1) and the conjecture cannot both hold. This is a conditional incompatibility, not a disproof of (5.1). Under the conjecture, Cauchy–Schwarz even forces $\sum_r |B_r(U)|^2 \gg U^3/\log^3 U$.

A weaker condition is enough. Write $k(U)$ for the number of lines $5 \le r \le U$, $E(U) = \sum_{r\le U} |B_r(U)|^2$ and

$$Q(U) = \frac{k(U)E(U)}{M(U)^2}.$$

Cauchy–Schwarz and Theorem 3 give, with no assumption,

$$T(U) \ge M(U)(1 - \sqrt{Q(U)}).$$

So if $Q(U) \le \eta < 1$ for some fixed η and all large U, there are infinitely many twin primes. This allows an energy of size $U^3/\log^3 U$, provided its constant is small enough. It is a sufficient condition only. No bound on Q is proved here. Cauchy–Schwarz and conditions of this shape are standard; nothing new is claimed for them.

U	1200	10^4	2×10^4	5×10^4	10^5
T/M	0.805010	0.795771	0.794442	0.793829	0.793733
Q	0.063546	0.074875	0.079064	0.085126	0.090695

(Regenerated by `code/verify_weighted_l2.py`*, which also checks* $\sum_r B_r = T - M$ *and checks* T *against an ordinary sieve of the twin pairs.)* Q stays far below 1 on this range, but it rises on every step from $U = 1200$. The data therefore do not show that Q stays bounded below 1.

The quantity in (5.1) is therefore the nested one of Theorem 3, not the raw strike deviation D_r below; the two are measured side by side in §5.3, and only the first is tied to $T - M$ by an identity.

5.2 Why $X \asymp U^2$ is the critical scale

The available sector window at depth U has length $\asymp U^2$, while the full network cycle has length $\asymp e^U$. Theorem 4 establishes orthogonality **on the full cycle**; the question is whether it survives restriction to a window exponentially shorter.

5.3 Numerical stress test of the bound

U	lines	$\sum \lvert D_r \rvert^2$	$X + U^2$	ratio	printed in earlier versions	deterministic term alone
300	60	5.36×10^2	1.05×10^5	**0.0051**	1.68	1.64
600	107	2.23×10^3	4.20×10^5	**0.0053**	3.00	2.96
1,200	194	8.56×10^3	1.68×10^6	**0.0051**	5.48	5.44

With the birth of each line accounted for, the ratio is flat, and the growth reported in earlier versions is the deterministic term of §5.1, which supplies 98 per cent of those numbers at every U. The restricted-depth row of earlier versions was flat for the same reason and is withdrawn with the main row. Regenerated by `code/verify_l2_nested.py`.

And the flat row does not settle (5.1) either, because it is not the quantity (5.1) needs. The strikes counted in D_r are the raw ones; the error of Theorem 3 is built from the deviations $\varepsilon_r(u)$ of the strikes of r **on the survivors of the smaller lines**, and it is those that satisfy $\sum_r B_r = T - M$ exactly. There is no identity making $\sum_r D_r$ the error of the survivor count, so a bound on $\sum_r |D_r|^2$ carries no consequence for $T(U)$. Measured on the same sectors, $\sum_r |\sum_u \varepsilon_r(u)|^2$ divided by $X + U^2$ is 0.035, 0.040, 0.063 at $U = 300, 600, 1200$, an effective exponent of 2.43 — which separates neither a power nor $U^2 \log^A U$ on a range this short.

The L^2 experiment as run does not decide the bound in either direction.

5.4 Why: the nested masks

Δ_r is not a free difference: it is the quadratic motion seen **through the mask of all smaller lines**, $A_{r^-} = A_{\{5,\dots,r^-\}}$, so the sequence being tested changes with r. That is the reason the nested row of §5.3 is the relevant one, and the reason a bound for it does not follow from the full-cycle orthogonality of Theorem 4; the measurements above neither establish the nesting cost nor rule it out.

A large-sieve-type inequality adapted to such *nested* masks would be the kind of estimate needed to bound $Q(U)$. No such estimate is proved in this paper.

5.5 A target from earlier versions

Precisely a bound

$$|S_r(V)| \le C\sqrt{N_r(V)}\log^A U,$$

where $S_r(V)$ is the centred partial sum since the line's birth and $N_r(V)$ the number of sectors observed. Earlier versions called this weaker than Hardy–Littlewood. **That was wrong in the same way as (5.1).** Summed over the $k(U) \asymp U/\log U$ lines, a bound of this shape for every line gives $|T - M| = O(U^{3/2}\log^{A'} U) = o(M)$, so it too would force $T/M \to 1$, against the conditional limit $e^{2\gamma}/4$. Read as a target, it is replaced by the condition on $Q(U)$ in §5.1.

The tested data do not support a small uniform constant: the worst ratio $\max |S_r|/\sqrt{N_r}$ rises from 4.25 at $U = 4199$ to 5.75 at $U = 7199$. This does **not** refute the existence of some larger eventual constant. The polylogarithmic form is **beyond what computation can settle here**: each sector costs $O(u)$ work, so distinguishing $\sqrt{N}\log^A$ from $\sqrt{N}\log^{A+1}$ would require U in the millions.

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6. Transferring the depth distribution to a window

The question is whether the depth distribution of [P3, Thm 2] transfers from cycle to short window more stably than the raw survivor count. **On the tested ranges it does for soft weights.** The deterioration is not perfectly monotone — there is a small overshoot near $t = 3$ — but the sharp depth-zero endpoint is clearly separated, with ratio near 0.80.

Setup. Moving depth: each sector (u^2, v^2) is sieved by all lines $\le u$, and each cell is recorded with its state and its inheritance depth j. The comparison is against the exact cycle distribution implied by [P3, Thm 2] — each line contributes nothing with probability $(q-2)/q$, a lower-rail strike with probability $1/q$, an upper-rail strike with probability $1/q$. Four bands of u, up to 1.3×10^6 cells per band.

Raw ratios. Window over cycle: NN gives 0.80, NO/ON gives 0.914, OO gives 1.063 — the deficit of §3.2 seen state by state. The total-variation distance between the window and cycle depth *shapes* within a state is 0.106 (NO/ON) and 0.089 (OO), i.e. the shape already transfers about twice as well as the count.

The curve. Transfer ratio for the weight $\max(0, 1 - j/t)$:

t	400–800	800–1600	1600–3200	3200–6400
12 (linear on this range)	1.0000	1.0000	1.0000	1.0000
8	0.9998	0.9997	0.9996	0.9994
5	0.9979	0.9973	0.9975	0.9979
3	1.0176	1.0200	1.0209	1.0214
2.5	1.0082	1.0081	1.0083	1.0079
2	0.9832	0.9753	0.9724	0.9684
1.5	0.9528	0.9484	0.9461	0.9424
≤ 1, i.e. the indicator $[j = 0]$	**0.7974**	**0.8056**	**0.8016**	**0.7953**

Two structural remarks, both of which the table makes visible.

1. **Linear weights measure** 1.0000 **to the reported precision on these windows.** Structurally, each line contributes its cyclic mean up to an $O(1)$ boundary term, so linear statistics are expected to be much more stable than sharp indicators; the finite-window equality is empirical here, not an exact theorem.
2. **The ratio is nearly flat in** u for the softer tested weights, drifts by about -0.015 at $t = 2$, and is flat again near the 0.80 endpoint. The dependence on t is not strictly monotone because of the small overshoot around $t = 3$.

And at Richert's own parameters the transfer is essentially perfect. Taking $\mathcal{A}$ to be the integers $6b \pm 1$ in (u^2, v^2), sieving by the lines below $z = X^{1/8}$ and weighting the primes in $[z, y)$ with $y = X^{1/3.17}$ and $\lambda = 0.83$: the survivor-count ratio measures 1.0000 and the weighted-sum ratio 0.9998, flat across $u = 300$ to 4800. The reason is that at $k_1 = 8$ the sieve variable is $s = (1/2)/(1/8) = 4$, comfortably above 2 — Campbell [1] runs at $s = 3.33$. **The window-transfer deficit does not bite at those parameters at all; the bottleneck in the almost-primes problem is the numerical optimisation, not the transfer.**

> **Proposition 1 (measured).** On the tested moving-depth windows, linear functions of inheritance depth give transfer ratio 1.0000 to the reported precision. For every **tested** value $t \geq 1.5$, the weight $\max(0, 1 - j/t)$ stays within about 6% of 1 (within about 3.2% for the tested $t \geq 2$), while for $t \leq 1$ — exactly the indicator $[j = 0]$ — the measured ratio is approximately 0.80.

A second family: pattern transfer at fixed depth. Proposition 1 measures the transfer of a *weight* on one cell. The cycle law of [P3, §6.3] is more general than that: it transports an arbitrary finite pattern H of cells, with $N_{\text{new}}(H) = (q - \nu_q(H))N_{\text{old}}(H)$. Nothing above tests it beyond $|H| = 2$, and the patterns are the sharpest indicators the framework has — a pattern of k cells is a condition on $2k$ integers at once. This paragraph tests that law directly.

Setup, and a constraint that appears immediately. Fix a height X and a depth $z = X^{1/u}$, sieve the whole range $6n \pm 1 \leq X$ by every line $\leq z$, and write $S_n = 1$ when both members of cell n survive. For a pattern $H = \{h_1 = 0 < \ldots < h_k\}$ set

$$J(H) = \frac{\Pr\left(S_{n+h_1} = \cdots = S_{n+h_k} = 1\right)}{\Pr(S_n = 1)^k}, \qquad L(H) = \prod_{5 \leq q \leq z} \frac{1 - \nu_q(H)/q}{(1 - 2/q)^k},$$

so that $J/L = 1$ is exactly the assertion that the cycle law of [P3, §6.3] survives the passage to a real window at depth z. Here $\nu_q(H) = \text{card}\{\pm 6^{-1} - h_i \bmod q\}$, and the pattern must be admissible:

consecutive cells $H = \{0, 1, 2\}$ already give $\nu_5 = 5$ and density zero, so the shortest admissible patterns are $\{0, 1\}$, $\{0, 1, 3\}$, $\{0, 1, 3, 5\}$, $\{0, 1, 3, 5, 6\}$ — the last a condition on ten integers spanning 38.

Measured, at $X = 4 \times 10^9$ (all 6.67×10^8 cells; the $\pm$ column is $1/\sqrt{\text{count}}$):

u	z	H	count	J	L	J/L	$\pm$
4	251	$\{0, 1\}$	1,688,188	0.3978	0.3978	**1.0000**	0.0008
4	251	$\{0, 1, 3\}$	142,416	0.4207	0.4206	**1.0002**	0.0026
4	251	$\{0, 1, 3, 5\}$	19,817	0.7337	0.7348	**0.9985**	0.0071
4	251	$\{0, 1, 3, 5, 6\}$	727	0.3374	0.3412	**0.9887**	0.0371
3	1587	$\{0, 1\}$	567,431	0.3973	0.3970	1.0007	0.0013
3	1587	$\{0, 1, 3\}$	27,527	0.4164	0.4179	0.9964	0.0060
3	1587	$\{0, 1, 3, 5\}$	2,167	0.7082	0.7255	0.9762	0.0215
3	1587	$\{0, 1, 3, 5, 6\}$	51	0.3601	0.3340	1.0780	0.1400
2	63,246	$\{0, 1\}$	85,884	0.4014	0.3969	1.0113	0.0034
2	63,246	$\{0, 1, 3\}$	1,703	0.4442	0.4175	1.0640	0.0242
2	63,246	$\{0, 1, 3, 5\}$	50	0.7280	0.7242	1.0053	0.1414

At the moving depth $u = 2$ the residual deviation shrinks with height. For $H = \{0, 1\}$, $J/L = 1.0311,\ 1.0160,\ 1.0113$ at $X = 10^8, 10^9, 4 \times 10^9$. Writing the k-tuple count with the integral $\int_2^X \mathrm{d}t/\log^{2k} t$ rather than $X/\log^{2k} X$, and the same for the normalising $\Pr(S)^k$, the leading $1/\log X$ corrections cancel and the first surviving term is $1 + 2k(k-1)/\log^2 X$, i.e. 1.008 for $k = 2$ and 1.025 for $k = 3$ at $X = 4 \times 10^9$; the measured 1.0113 and 1.0640 sit within one to two standard errors of those. **We report this as an account of the observed size, not as a derivation.**

> **Proposition 2 (measured).** On the tested range, the pattern law of [P3, §6.3] transfers to a real window at fixed depth with $J/L = 1$ to within the statistical error, for every tested admissible pattern of up to five cells — a condition on ten integers — at $u = 3$ and $u = 4$. At the moving depth $u = 2$ the ratio exceeds 1 by one to six per cent, decreasing with height.

What this adds to Proposition 1, and it changes the reading of this section. Proposition 1 shows the one-cell statistic losing about 20% exactly when the weight sharpens to the depth-zero indicator. Proposition 2 shows a far sharper indicator — the simultaneous survival of ten integers — transferring at 1.0000. **The two are consistent, and together they separate the loss into two factors: it lies entirely in the one-point density, which carries the Buchstab correction at $s = 2$ (§3.2), and not in the correlations built on top of it, which transfer exactly for every pattern tested.** In other words the deficit is a function of the depth relative to the window, not of the sharpness of the statistic.

What this settles, and what it does not. The experiment shows a sharp distinction inside the tested weight family: soft weights transfer well, whereas the depth-zero indicator carries the large deficit. That indicator is the twin condition in the moving square window. The experiment does not prove a transfer theorem for arbitrary weights or all t; it locates the observed loss in this family. Proposition 2 then locates it more precisely still — in the one-point density rather than in the pattern structure — but it too is a measurement on a finite range and proves no transfer theorem.

The transfer fails in the one-point density and holds in the correlations, for every pattern tested up to ten integers.

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7. What this paper establishes

Proved. Old lines act as rulers with phase $\phi_q(p) \equiv (q-p)/2$ and the binary decision for $q \geq 7$ (Theorems 1, 2); the telescoping identity reducing the iterative deviations to a single sum (Theorem 3); the full-cycle martingale structure with its energy identity (Theorem 4); the square-cycle cancellation for one line and for any set of lines (identity B), the mean law with its Legendre symbol over a full period (identity A), and its telescoping consequence (§4.3).

Measured, not proved. The stability of $T/M \approx 0.80$ and its agreement with the conditional limit $e^{2\gamma}/4$ (§3.2); the depth profile of §3.2 against its Buchstab model; the values of $Q(U)$ in §5.1; the state-space observation (§3.5); the two L^2 rows of §5.3; the transfer curve of §6 (Proposition 1); and the pattern transfer of §6 (Proposition 2). The data do not refute a polylogarithmic L^2 bound asymptotically, and the finite-window value 1.0000 for linear weights is a measurement to the reported precision, not an exact equality theorem.

The shape of the result. On the tested windows, soft weights preserve the cycle prediction far better than the sharp depth-zero indicator. The latter is precisely the primality/twin atom in this setup.

The exact cycle laws are proved; the short-window transfer remains partly empirical.

The applications of that statement, and the problems it does and does not decide, are Papers 5 to 8.

No progress toward the twin-prime conjecture is claimed, and no new bound. Priority is not claimed for any result.

The computations and much of the prose in this paper were prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running the computations, and for auditing the papers against their own scripts. All statements were checked by the author, who is responsible for them; the repository README sets out the division of labour in full.

References

The companion papers of this set are cited as [P1] to [P12], and the numbered entries below are the external works. The two kinds never share a number: a bracket with a P is a companion paper, a bare number is a reference in the list below. This paper imports only the definitions and statements of the companion papers, never their proofs.

1. P. Campbell, *On the existence of integers with at most 3 prime factors between every pair of consecutive squares*, arXiv:2603.10356 (2026). — *the almost-primes computation whose parameters are used in §6; it runs at sieve variable* $s = 3.33$.
2. J. Friedlander and H. Iwaniec, *Opera de Cribro*, AMS Colloquium Publications **57**, 2010. — *for the Buchstab-type correction at* $s = 2$ *discussed in §3.2, and for the large-sieve inequalities of the kind §5.4 would need.*
3. G. H. Hardy and J. E. Littlewood, *Some problems of 'Partitio Numerorum'; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70. — *for the conjectured twin-prime count behind the conditional limit* $e^{2\gamma}/4$ *of §3.2 and §5.1.*